

Functional properties of the titin/connectin-associated proteins, the muscle-specific RING finger proteins (MURFs), in striated muscle

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Abstract

The efficient functioning of striated muscle is dependent upon the proper alignment and coordinated activities of several cytoskeletal networks including myofibrils, microtubules, and intermediate filaments. However, the exact molecular mechanisms dictating their cooperation and contributions during muscle differentiation and maintenance remain unknown. Recently, the muscle specific RING finger (MURF) family members have established themselves as excellent candidates for linking myofibril components (including the giant, multi-functional protein, titin/connectin), with microtubules, intermediate filaments, and nuclear factors. MURF-1, the only family member expressed throughout development, has been implicated in several studies as an ubiquitin ligase that is upregulated in response to multiple stimuli during muscle atrophy. Cell culture studies suggest that MURF-1 specifically has a role in maintaining titin M-line integrity and yeast two-hybrid studies point toward its participation in muscle stress response pathways and gene expression. MURF-2 is developmentally down-regulated and is assembled at the M-line region of the sarcomere and with microtubules. Functionally, its expression is critical for maintenance of the sarcomeric M-line region, specific populations of stable microtubules, desmin and vimentin intermediate filaments, as well as for myoblast fusion and differentiation. A recent study also links MURF-2 to a titin kinase-based protein complex that is reportedly activated upon mechanical signaling. Finally, MURF-3 is developmentally upregulated, associates with microtubules, the sarcomeric M-line (this report) and Z-line, and is required for microtubule stability and myogenesis. Here, we focus on the biochemical and functional properties of this intriguing family of muscle proteins, and discuss how they may tie together titin-mediated myofibril signaling pathways (perhaps involving the titin kinase domain), biomechanical signaling, the muscle stress response, and gene expression.

Introduction

The efficient functioning of striated muscle cells is dependent upon the precise interactions and alignments of complex cytoskeletal networks. For example, sarcomeres, the basic contractile units of myofibrils, are comprised of uniformly arranged filament systems and regulatory proteins. The actin-containing thin filaments are anchored in the Z-lines and extend toward the middle of the sarcomere, the M-line, where they interact with the myosin-containing thick filaments to drive contraction. A third filament system is composed of single molecules of titin, the largest vertebrate protein identified to date (M_r 3–3.7 MDa), also referred to as connectin (Maruyama *et al.*, 1977; Wang *et al.*, 1979). Titin filaments span half sarcomeres, with their N-termini (~80 kDa region) overlapping in the Z-lines and their C-termini (~200 kDa region) overlapping in the M-lines, thus forming a continuous filament system between adjacent myofibrils (Obermann *et al.*, 1997; Gregorio *et al.*, 1998; Mues *et al.*, 1998). Based

on titin's assembly properties, molecular layout and modular structure, it is proposed to act as a template for myofibrillogenesis. Additionally, titin's I-band region contains multiple elastic elements that contribute to muscle stiffness (for reviews see: Linke and Granzier, 1998; Tskhovrebova and Trinick, 2003; Granzier and Labeit, 2002, 2004). Finally, titin contains several phosphorylation motifs, calpain-binding sites and its C-terminal (M line) region contains a serine/threonine (Ser/Thr) kinase domain with no known physiologic catalytic function. Excitingly, recent studies have identified various signaling molecules associated with the giant molecule including the muscle RING finger (MURF) proteins; muscle ankyrin-repeat proteins (CARP, ankrd2, and DARP); muscle-specific calpain, p94; and obscurin (Labeit and Kolmerer, 1995; Sorimachi *et al.*, 1995; Bang *et al.*, 2001; Centner *et al.*, 2001; Young *et al.*, 2001; Miller *et al.*, 2003, 2004). Thus, this giant protein has multiple roles in muscle, one of which may include regulating various signaling pathways involved in myofibril turn-over, assembly, and/or muscle gene regulation. The focus of this article is to discuss the potential functional roles of the unique family of titin-associated proteins, the MURFs.

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Structure of the MURF family

The titin kinase domain is surrounded by immunoglobulin (Ig) and fibronectin (Fn)-like domains at the M-line region of the sarcomere, and this entire region is highly conserved in vertebrate titin, in various invertebrate titin-like proteins including projectin and twitchin, and in myosin light chain kinase (MLCK) (Benian *et al.*, 1989; Olson *et al.*, 1990; Southgate *et al.*, 1991; Labeit *et al.*, 1992; Flaherty *et al.*, 2002; for reviews of titin Ig and Fn domains see: Bullard *et al.*, 2002; Tskhovrebova and Trinick, 2004). However, the exact function of this domain remains elusive. Yeast-two hybrid screens aimed at identifying potential substrates and regulators of this region identified MURF-1 as a ligand of titin's A168/A169 Ig motifs, located just N-terminal to the kinase domain at the periphery of the M line (Centner *et al.*, 2001). Consistent with these *in vitro* binding data, MURF-1 assembles at the M-line region of many myofibrils, in addition to the Z-line in some muscles (Centner *et al.*, 2001; McElhinny *et al.*, 2002). Additionally, MURF-1 was described as striated muscle RING zinc finger (SMRZ), detectable in some nuclei (Dai and Liew, 2001; McElhinny *et al.*, 2002). When MURF-1 was used as bait in yeast two-hybrid studies, a protein with 62% homology to MURF-1 was identified as a binding partner and named MURF-2. Further, MURF-2 baits revealed a third family member, corresponding to a protein previously described as "MURF" (Spencer *et al.*, 2000), sharing 77% homology with MURF-1 and 65% homology with MURF-2. This protein was referred to as MURF-3, since it was the third MURF family member identified in the yeast two-hybrid screens. Additional studies determined that the MURF proteins were products of three different genes with distinct expression patterns (summarized in Table 1), and that MURF-1 was the only family member that appeared to interact directly with titin. However, all MURFs can homo- and hetero-oligomerize, likely via their coiled coil domains (Spencer *et al.*, 2000; Centner *et al.*, 2001; McElhinny *et al.*, 2002). Therefore, we hypothesize that MURF-2 and MURF-3 may associate indirectly with titin, via MURF-1.

Structurally, all MURFs (also known as RNF 28, 29, and 30, respectively) share a highly conserved N-terminal RING domain followed by a MURF family-specific region, a Zn-binding B-box domain, leucine-rich coiled-coil motifs, and an acidic tail (Spencer *et al.*, 2000; Centner *et al.*, 2001; Dai and Liew, 2001) (Figure 1). Therefore, the MURFs are members of a growing family of RING-B-box-coiled-coiled (RBCC) proteins, molecules implicated in diverse biological functions such as signal transduction, gene transcription, differentiation, ubiquitination and morphogenesis (for reviews see: Borden, 1998; Freemont, 2000). In fact, it is proposed that all RING domain-containing proteins are involved in ubiquitination, likely acting as E3 ligases to regulate protein degradation (Freemont,

2000), and/or as scaffolds for the formation of large protein complexes (Borden, 2000). These reports strongly suggest that MURFs have multiple physiological roles in muscle, an idea that we discuss here.

Materials and methods

Cell culture, transfection, and antisense oligonucleotide treatment

For skeletal myotube cultures, myoblasts were isolated from day 11 chick embryo pectoralis muscle as described (McElhinny *et al.*, 2004). Cardiac myocytes were isolated from day 6 embryonic chick hearts or fetal rats and maintained as described (Gustafson *et al.*, 1987; Gregorio and Fowler, 1995). Isolated cells were plated in 35 mm tissue culture dishes for RT-PCR analyses, or in dishes containing 12 mm round coverslips for immunofluorescence staining ($\sim 10^6$ cells/dish). 15–20% of the cells in our primary cultures are fibroblasts. Rat cardiac myocytes were treated on day 3 with MURF-3 antisense oligonucleotides. 10 μ g of control or antisense oligonucleotides were added to 7 μ l of Cytofectene (Biorad) and the DNA-lipid complex was incubated in 200 μ l OptiMEM for 15 min at 37°C. Our transfection efficiency was >80%. Three, 16-mer antisense oligonucleotides were designed using the mouse MURF-3 sequence. Two of the antisense oligonucleotides knocked down endogenous chick MURF-3 levels: 5'-TGT GCC GCA AAT GTG C-3' and 5'-ATG-GAG-GAG-GTG-TGC-C-3'. For control oligonucleotides, reverse (but not complementary) sequences to the oligonucleotides listed above were used. For expression of human GFP-MURF-3, full-length MURF-3 cDNA was amplified by PCR with a S25/S26 primer pair, with *EcoRI* and *BamHI* restriction sites: F=TTTGAATTCATGGACAACCTGGAGAA GCAGCTCATCTGCCC and R=TTTGGATCCGG GCTCTCGAGCAGGGTCGGGTCGGCGCAGGC (see www.titin.org). The 1.2 kb inserts were cloned into pEGFP-C2 (Clontech) and 1 μ g was transfected into chick cardiac myocytes.

RT-PCR analyses of MURF-3 mRNA transcripts

For amplification of MURF-3 mRNA transcripts, rat cardiac myocyte cultures were harvested for total RNA 24, 48 and 72 h after treatment with oligonucleotides using a RNeasy Mini kit (Qiagen). cDNA was prepared from 20 μ l of total RNA using the Super Script-II Reverse Transcriptase (Invitrogen), and the cDNA concentration was determined using an Eppendorf Biophotometer. 50–100 ng of template were used per PCR reaction. Rat MURF-3 primers were used for amplification in 25 μ l reactions using Qiagen PCR kits. The primers used were MURF-3 F 5' GAA CTT CAC GGT GGG TTT CAA G 3' and R 5' CAC TCT

Table 1. MURFs expression patterns, localizations and functions

	Expression patterns	Localization	Proposed functions
<i>MURF-1</i>	Throughout development Up-regulated in response to stress/atrophy signals	Myofibrils (Z disc & M line) Nucleus and peri- nuclear envelope	Binds: A168-A170 of titin SUMO-3 GMEB-1 ISOT-3 Ubc-9 ATP synthase Creatine kinase Nebulin N-RAP TnT MLC-2 Myotilin T-cap (telethonin) Cardiac troponin I Receptor for activated protein kinase C (RACK1) MURF-1, 2 and 3 Ubiquitin ligase activity Required for thick filament stability and titin's M line structure Degradation of proteins during skeletal muscle atrophy Important role in myofibril maintenance and turnover Regulator of the PKC-dependent cardiac hypertrophic response Regulator of energy metabolism?
(Centner <i>et al.</i> , 2001; Bodine <i>et al.</i> , 2001; Dai and Liew, 2001; McElhinny <i>et al.</i> , 2002; Gotthardt <i>et al.</i> , 2003; Arya <i>et al.</i> , 2004; Kedar <i>et al.</i> , 2004; Lecker <i>et al.</i> , 2004; Nikawa <i>et al.</i> , 2004; Witt <i>et al.</i> , 2005)			
<i>MURF-2</i>	Down-regulated during development	Stable microtubules (Glu-MTs) Myofibrils (M line) Nucleus (upon serum starvation or mechanical inactivity)	Binds: MURF-1,-2,-3 M line titin? P62/SQSTM1 Nebulin N-RAP TnT MLC-2 Myotilin T-cap (telethonin) Proposed adaptor between microtubules and titin/myofibrils during assembly Critical for myoblast fusion and myofibrillogenesis Required for stability of M-line titin and other M-line proteins Critical for stability of stable MTs and intermediate filaments Proposed member of titin kinase-based signaling complex to sense mechanical load
(Centner <i>et al.</i> , 2001; Pizon <i>et al.</i> , 2002; McElhinny <i>et al.</i> , 2004; Lange <i>et al.</i> , 2005; Witt <i>et al.</i> , 2005)			
<i>MURF-3</i>	Up-regulated during development	Glu-MTs Myofibrils (Z disc & M line)	Binds: MURF-1,-2,-3 Microtubules Establish/maintain stable Glu-MT array Required for skeletal myoblast fusion, initiation of myogenesis, gene expression, and differentiation
(Spencer <i>et al.</i> , 2000; Centner <i>et al.</i> , 2001; this study)			

TCT GGC GTT TGT AAA TGG 3', which amplify a 515 bp product. An annealing temperature of 60°C was used for MURF-3 primers, and 25 cycles were performed. Rat glyceraldehyde-3-phosphate dehydrogenase (GAPDH) also was amplified as a loading control (using F 5'-CCAGTATGATTCTACCCACGG C-3' AND R 5'-CGGAGATGATGACCCTTTTGG C-3'), which amplified a 227 bp product. All PCR products were sequenced for verification.

Indirect immunofluorescence microscopy

Cells were stained essentially as described previously (Gregorio *et al.*, 1998), except that the cells used in the antisense experiments were fixed in methanol at -20°C for 10 min. Cells were incubated with goat anti-MURF-3 (ab4351) antibodies (10 µg/ml; Abcam: Cambridge, UK), followed by Texas Red-conjugated rabbit anti-goat IgG (1:600; Abcam). Next, the cells were costained

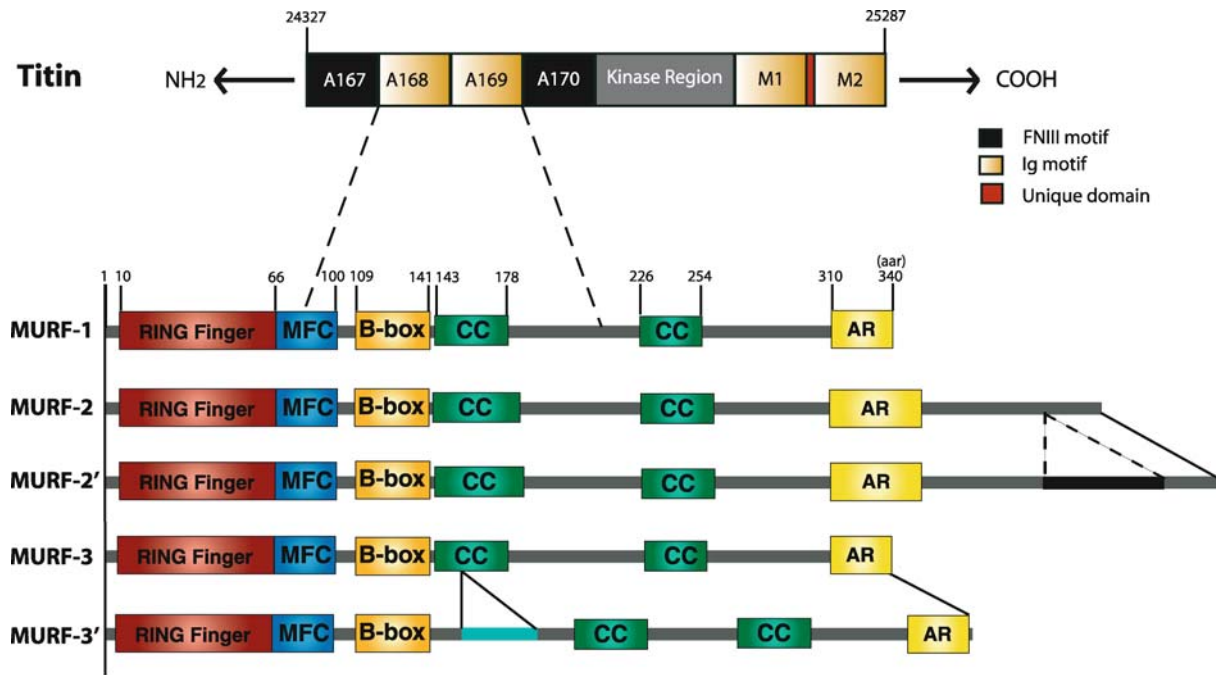


Fig. 1. Schematic of the Domain Structure of the MURFs. (a). Structurally, the MURFs contain a RING-finger domain (red), a MURF family conserved region (blue), a B-box domain (yellow), two coiled-coil domains (green), and an acidic tail (yellow). The C-terminal acidic residue-rich domains within the MURF-family are more divergent and also subject to differential splicing events. This figure was modified from one first presented in Centner et al. (2001). MURF-1 was first identified as an interacting partner of titin A168-A170 (shown above), which is located ~200 residues (~10 nm) amino-terminal to the titin kinase domain at the periphery of the M line lattice. mRNA studies by Pizon et al. (2002) reported two additional MURF-2 isoforms generated by differential splicing events: a p27 isoform, missing the coiled coiled domains and p60B, containing an alternate C-terminus generated by translation of alternative frames in the terminal exon (not shown).

with AlexaFluor 488-conjugated phalloidin (Molecular Probes). The anti-MURF-3 antibody used in this study was generated against the peptide GSAGPEEERPDGP from the C-terminus of the molecule (Abcam). The specificity of the antibody was tested by ELISA and Western blot analysis by the company, where a single band at 45 kDa was detected in human muscle lysates (see: www.abcam.com/?datasheet=4351). Cells transfected with GFP-MURF-3 were stained with anti-sarcomeric α -actinin (1: 1:500) (EA-53, Sigma-Aldrich) antibodies, followed by Texas Red-conjugated goat anti-mouse IgG (1:600) (Jackson ImmunoResearch Laboratories). Cardiac myocytes treated with control or MURF-3 antisense oligonucleotides were fixed 48 h after treatment and stained with anti- β -tubulin (clone TUB 2.1) antibodies (Sigma-Aldrich) and Alexafluor 488-conjugated goat anti-mouse IgG, followed by rabbit anti-titin A168–170 (at 1:200) antibodies (Centner et al., 2001) and Texas Red-conjugated goat anti-rabbit IgG. Coverslips were mounted on slides using Aqua Poly/Mount (Polysciences, Inc.) and subsequently analyzed on a Zeiss Axiovert microscope using a 100 $\times$ objective (NA 1.25), and micrographs collected as digital images on a Hamamatsu Orca-ER camera using OpenLab software (Improvision, Lexington, MA) (Figure 3). Images were also collected on a DeltaVision Deconvolution Model D-OL Olympus microscope with a 100 $\times$ objective (1.3 NA) using a Photometrics Series 300 CCD camera (Figure 2d, e) and on a Zeiss 510 multiphoton microscope using a 100 $\times$ (1.4 NA)

objective (Figure 2a–c). Images were processed for presentation using Adobe Photoshop 7.0.

Results and discussion

MURF-1: A link between myofibril turn-over and gene expression?

Initial evidence for MURF-1's physiological function came from transcript profiling studies that classified it among a small subset of molecules up-regulated in multiple models of skeletal muscle atrophy, a complex phenomenon involving muscle protein degradation in response to many stimuli including: denervation, immobilization/injury, glucocorticoid treatment, space flight and aging (Bodine et al., 2001). Consistent with these data, recombinant MURF-1 exhibited ubiquitin ligase activity and MURF-1 null mice were resistant to atrophy after skeletal muscle denervation (Bodine et al., 2001). Several subsequent studies confirmed that MURF-1 levels were significantly upregulated in other models of muscle atrophy including fasting, as well as in rats with cancer cachexia, streptozotocin-induced diabetes mellitus or uremia induced by subtotal nephrectomy, and in rats injected with lipopolysaccharide (LPS) (Dehoux et al., 2003; Lecker et al., 2003, 2004; Nikawa et al., 2004). Together, the results from these studies indicate that MURF-1 is a component of multiple parallel pathways involving the ubiquitin-pro-

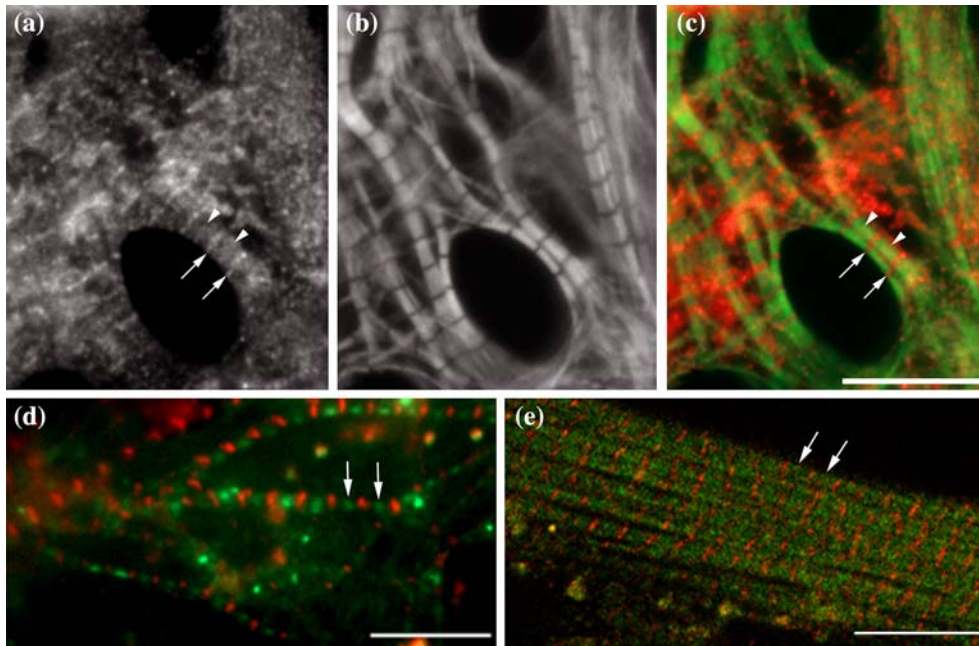


Fig. 2. MURF-3 is localized to the M-line in some myofibrils in cardiac and skeletal myocytes. (a) Day 5 rat cardiac myocytes were fixed and stained for MURF-3 (a, red in c) and fluorescently-conjugated phalloidin (b, green in c). Staining for MURF-3 was detected diffuse in the cytoplasm, and predominantly at the M line in some myofibrils (long arrows); Z line staining was also visible (short arrows). Chick cardiac (d) and skeletal (e) myocytes transfected with GFP-MURF-3 (green) demonstrate assembly of the fusion protein at the M-line (arrows); staining for the Z line marker α -actinin in red. The significance of the broad pattern of GFP-MURF-3 localization in skeletal myocytes, in comparison to its more narrow distribution in cardiac myocytes, is unclear. Bars = 10 μ m.

teasome system that is activated during skeletal muscle atrophy. Therefore, muscle mass is likely maintained by balancing the levels and/or activities of a subset of molecules including MURF-1.

Recent studies have begun to unravel the complex molecular mechanisms by which the expression of MURF-1, and other specific ubiquitin ligase “atrogenes” (such as atrogin-1) are regulated (see Franch and Price, 2005 for review). First, insulin-like growth factor-1 (IGF-1), a key player in myocyte differentiation and growth, suppresses transcription of MURF-1 and atrogin-1 (Sacheck *et al.*, 2004). This inhibition of MURF-1 and atrogin-1 occurs via the phosphatidylinositol 3-kinase/Akt/mammalian target of rapamycin (PI3K/Akt/mTOR) pathway (Latres *et al.*, 2005). Specifically, it was determined that this inhibition of the two ubiquitin ligases involves Akt-mediated inhibition of the FOXO family of transcription factors (Stitt *et al.*, 2004). These studies have led to the hypothesis that at least some of the molecular pathways regulating atrophy and hypertrophy are not separate, but linked. Additionally, a separate pathway appears to be activated during atrophy. Transgenic mice expressing an activated form of IkappaB kinase beta (MIKK) exhibit profound muscle wasting and an increased expression of MURF-1 (Cai *et al.*, 2004). All of these studies have laid the foundation for the treatment of muscle atrophy via targeting of these pathways.

Mechanistic insight into exactly how MURF-1 functions was obtained from the identification of its binding partners. First, MURF-1’s RING domain interacts

with SMT3b/SUMO-2 (small ubiquitin-related modifier), a member of an ubiquitin-related family of proteins (Dai and Liew, 2001). MURF-1 also binds to ubiquitin-conjugating enzyme-9 (Ubc-9) and isopeptidase T-3 (ISOT-3), enzymes involved in SUMO-modification (SUMOylation) and ubiquitin-hydrolysis, respectively (McElhinny *et al.*, 2002). Intriguingly, although ubiquitination and SUMOylation of proteins involve structurally similar pathways, their functional implications are distinct. SUMOylation does not appear to target proteins for proteasome-based degradation, but regulates diverse processes including nuclear import, gene expression, and subcellular targeting (for reviews see Melchior, 2000; Melchior *et al.*, 2003; Gill, 2004). Thus, if MURF-1 is indeed sumoylated, it is tempting to speculate that this process mediates its targeting to titin and/or the nucleus. Alternatively, MURF-1 may not be sumoylated, but may instead interact with other proteins that are sumoylated; these potential mechanisms need to be clarified. In any case, certain SUMO proteins are components of cellular stress response pathways (Saitoh and Hinchee, 2000), and so MURF-1’s interaction with SUMO may also mediate its activities in response to atrophy stimuli. In support of its proposed role in these pathways, MURF-1 appears to be the only MURF member that binds to the transcriptional regulator, glucocorticoid-modulatory element binding protein-1 (GMEB-1) (McElhinny *et al.*, 2001), and MURF-1 mRNA levels increase in response to glucocorticoid exposure, a treatment known to cause atrophy (Bodine *et al.*,

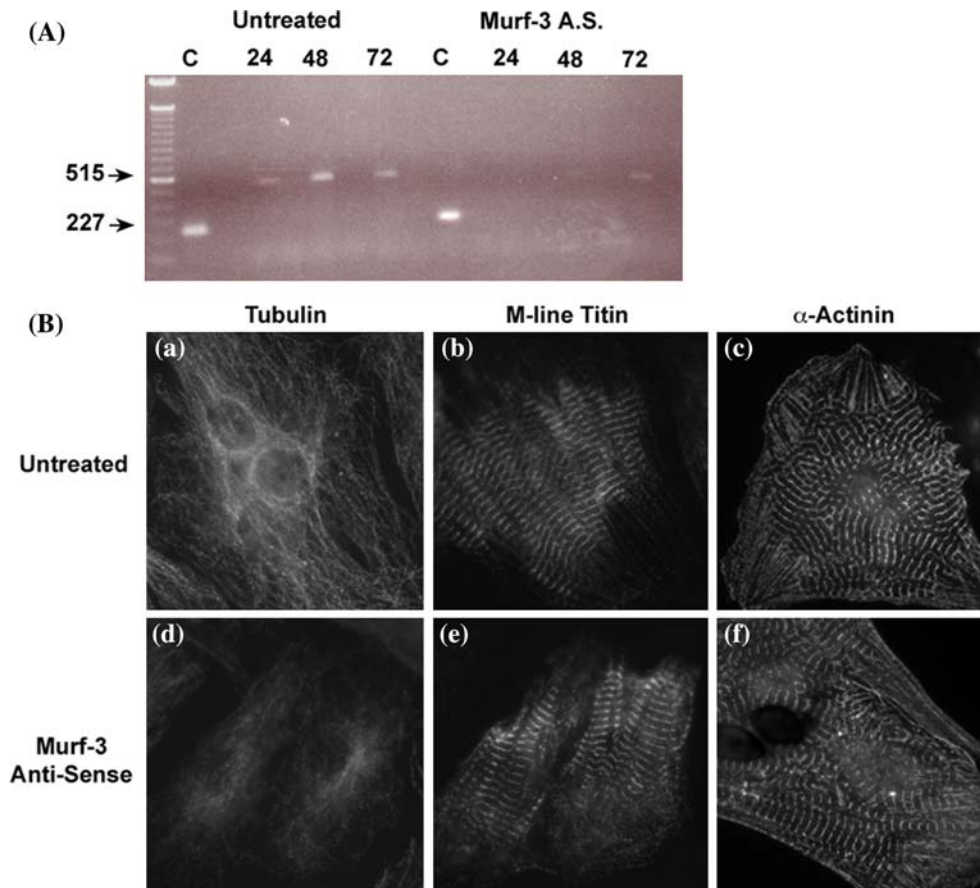


Fig. 3. Microtubules are specifically perturbed when MURF-3 levels are knocked down in cardiac myocytes. (A) Cardiac myocytes were treated with MURF-3 antisense or control oligonucleotides and RT-PCR was performed to analyze mRNA transcript levels. MURF-3 mRNA transcript levels (515 bp product) were significantly decreased in cultures treated with MURF-3 antisense oligonucleotides (right side of gel) compared to control oligonucleotides (left side of gel). GAPDH (227 bp product) was amplified as a loading control. (B) Day 3 rat cardiac myocytes were treated with control oligos (a–c) or MURF-3 antisense oligos (d–f) and incubated for 48 h before fixation. Staining for α -tubulin, which labels all microtubule populations, appeared in a complex, radial web-like array in control myocytes (a), but appeared significantly sparser in MURF-3 antisense-treated cells (d). M line titin (e), α -actinin (f) and thin filaments (not shown) were not affected by MURF-3-antisense treatment, in comparison to controls (b, c). Note, cells in a and b, and in d and e were double stained. At 72 h post-treatment, α -tubulin started to recover in its structural integrity in the MURF-3 antisense treated cells and within 96 h of treatment, appeared similar to controls (not shown). Bar = 10 μ m.

2001). Further studies are required to determine whether MURF-1 acts as a direct link between titin and gene expression.

How might MURF-1's proposed functions involve titin? Initial clues were obtained from functional studies in primary cultures of chick cardiac myocytes. Over-expression of GFP-MURF-1, expression of its central region only (containing its titin A168–170-binding site), or expression of GFP-titin A168–170 resulted in a significant and specific perturbation of the structure of titin's M-line region (McElhinny *et al.*, 2002). In turn, thick filament components also were disrupted, but not other regions of titin, the Z-lines, or the thin filaments. Thus, the interaction of titin with MURF-1 may regulate and/or stabilize the structure of the M-line region of the sarcomere. Normal expression of MURF-1 could, therefore, be essential for mediating myofibril turn-over/assembly throughout muscle development. In light of the role of MURF-1 in muscle atrophy, it is possible that MURF-1

functions in titin-based degradation/ubiquitination pathways, leading to muscle atrophy.

Recent experiments in neonatal rat cardiomyocytes reveal that MURF-1 acts as an ubiquitin ligase that targets cardiac troponin I for degradation in a proteasome-dependent manner. It was proposed that the association of MURF-1 with troponin I may “destabilize” sarcomeres upon binding to titin, thus allowing MURF-1 access to troponin I (Kedar *et al.*, 2004). Further investigations by the same group determined that MURF-1 also interacts with the receptor for activated protein kinase C (RACK1), and the two molecules colocalize in rat myocytes after treatment with phenylephrine (PE), an α -adrenergic receptor agonist known to induce a hypertrophic response in cultured cells (Arya *et al.*, 2004). Furthermore, MURF-1 inhibits PKC ϵ translocation to focal adhesions, blocking their formation, and also inhibits the activity of the downstream signaling molecule, ERK1/2. Strikingly, MURF-1 also abrogated PE-induced cellular hypertro-

phy. Taken together, these data suggest that MURF-1 regulates hypertrophic pathways involving PKC and may in fact, be required to prevent cardiac hypertrophy. Exactly how MURF-1's multiple binding partners and their emerging potential pathways (i.e., ubiquitination, sumoylation, steroid-regulation, PKC activity, the IGF-1-based pathway, and cellular atrophy/hypertrophy) are involved in titin-based myofibril assembly and turn-over processes remains to be determined.

Finally, elegant experiments using conditional mouse models expressing truncated titin molecules (missing exons MEx1–2, which encode ~200 kDa of titin's M-line region, including the kinase domain and the MURF-1 binding region) also may point to a role for MURF-1 in M line structure. Early expression of the mutant titin was embryonic lethal, while expression late in development resulted in gradual sarcomeric disassembly (starting with disruption of M-line titin), and muscle weakness, leading to a progressive myopathy and death at 5 weeks of age (Gotthardt *et al.*, 2003). Further studies are required to determine specifically the contribution of titin's MURF-1 binding site, in comparison to the contribution of titin's kinase domain, to this severe phenotype. Interestingly, microarray studies performed in the Gotthardt study also revealed that although MURF-1 levels were not affected by the titin M-line truncation, other titin-associated proteins CARP and ankrd2 were significantly up-regulated (Gotthardt *et al.*, 2003). This is particularly interesting since the levels of CARP, a negative regulator of transcription, and ankrd2 are known to be induced under other conditions of elevated stress or strain including injury and hypertrophy (CARP), and stretch or denervation (ankrd2/Arpp) (Jeyaseelan *et al.*, 1997; Zou *et al.*, 1997; Kemp *et al.*, 2000; Ikeda *et al.*, 2003). It is tempting to speculate that MURF-1 and CARP signaling pathways exhibit cross-talk that is mediated by the titin kinase region. For further discussion on the potential links between titin and biomechanical signaling see: Miller *et al.*, 2004; Granzier and Labeit, 2004.

MURF-3: The founding MURF family member has critical roles in microtubule stabilization and myogenesis. Is it also linked to titin?

The original MURF, known to be up-regulated during striated muscle development, was discovered as a putative binding partner for serum response factor (SRF) in yeast two-hybrid screens (Spencer *et al.*, 2000). Although this interaction could not be confirmed by the authors using other assays, their further characterization of MURF-3 certainly revealed very important findings and unique functions. First, based on MURF-3's homology with the human microtubule-binding Midline (Midl) proteins, it was investigated whether it shared this function. Indeed, co-sedimentation assays determined MURF-3 associated with microtubules via its RING and coiled coil domains, and immunofluo-

rescent studies found that it colocalized with microtubules in skeletal muscle, as well as detectable at the Z-line of sarcomeres (Spencer *et al.*, 2001). Furthermore, ectopic expression of MURF-3 in COS cells established a population of microtubules resistant to depolymerization. Consistent with these data, its association with microtubules led to the formation of Glutamic acid-modified microtubules (Glu-MTs), a specific post-translationally modified population "marked" with enhanced stability. Reduced levels of MURF-3 via antisense techniques in C2C12 myotubes inhibited Glu-MT formation and blocked myogenic gene expression, differentiation and myotube fusion (Spencer *et al.*, 2001).

What is the link between microtubule stability and myogenesis? Investigating the exact functions of microtubules in muscle is difficult, since their distinct arrays (i.e., stable vs. dynamic microtubules), and their organization varies between individual muscle cells in culture, among muscle types, developmental stages, and disease states (i.e., Saitoh *et al.*, 1988; Gundersen *et al.*, 1989; Kano *et al.*, 1991; Webster, 1997; Belmadani *et al.*, 2002). However, previous studies have shed some light on this issue, since perturbation of microtubule dynamics with depolymerizing or stabilizing agents blocked myoblast fusion, differentiation, and myofibrillogenesis in skeletal myocytes (Antin *et al.*, 1981; Toyama *et al.*, 1982; Saitoh *et al.*, 1988). Furthermore, it is clear that the establishment and maintenance of the stable Glu-MT array (the population with which MURF-3 associates) is a requirement for muscle differentiation, gene expression, and myotube fusion (Gundersen *et al.*, 1989; Chang *et al.*, 2002). A lingering critical question, however, is what mechanisms are responsible for the formation and maintenance of the stability of these distinct microtubule populations? The molecular identities of microtubule associated proteins (MAPs) specific for Glu-MTs are unknown and therefore, it is an exciting possibility that MURF-3 (and MURF-2, see below) may function as Glu-MT stabilizers in muscle, perhaps at different developmental stages.

Several questions remain concerning MURF-3. First, what is responsible for its association at the Z-line? One idea is that it may interact with MURF-1 at the Z-line, since a portion of MURF-1 is detected there in adult muscles (Centner *et al.*, 2001). Second, does MURF-3 also interact adjacent to the titin kinase domain (perhaps via other MURFs), and if so, is MURF-3 a link between titin and microtubules? Third, what is the function of MURF-3 in cardiac muscle? To begin to address the latter questions, we stained primary cultures of rat cardiac myocytes with polyclonal anti-MURF-3 antibodies from a commercial source (Abcam). Costaining with antibodies to the abundant Z line component, sarcomeric α -actinin, revealed that MURF-3 is predominantly localized to the M line in this cell type; however, Z line staining was detected in some myofibrils (Figure 2a). Further-

more, transfection of GFP-MURF-3 into chick cardiac and skeletal myocytes also revealed that MURF-3 assembled at the M line in some myofibrils in both cell types (Figure 2d, e), but interestingly, was never observed at the Z line (Figure 2). GFP-MURF-3 was also found associated with linearly aligned “filamentous” structures in skeletal myocytes via confocal microscopy (data not shown), consistent with it being reported as a microtubule-associated protein (Spencer *et al.*, 2001).

The prominent M-line-associated staining for MURF-3 is particularly interesting because it suggests that MURF-3 may also associate with titin A168/169, probably via heter-oligomerization with MURF-1 and/or MURF-2. As an initial approach to study the functional properties of MURF-3 in cardiac myocytes, antisense approaches were utilized. Specifically, reduction of MURF-3 mRNA transcripts via antisense methods in cardiac myocytes resulted in a significant perturbation of microtubules, indicating its importance for microtubule stabilization in cardiomyocytes (Figure 3). Strikingly, all myofibril components examined appeared unperturbed in structure. Why M-line staining for MURF-3 was detected in this study but not in the study by Spencer *et al.* (2001) is unclear, but it could be due to differences in muscle types, developmental stages and/or potential isoform differences. Furthermore, it is unclear why neither GFP-MURF-3 nor staining with anti-MURF-3 antibodies appeared to be associated with microtubules in the cardiac myocytes, yet reduction of its expression disrupted microtubule integrity. Further experiments are required to determine exactly what populations of microtubules are affected by MURF-3 knockdown and how MURF-3 affects microtubule structure in cardiac myocytes. This is especially important given that in heart, disruption of microtubules perturbs gene expression, Ca^{2+} signaling, myosin mRNA transport, and cellular viscosity (Perhonen *et al.*, 1998; Takahashi *et al.*, 1998; Gomez *et al.*, 2000; Calaghan *et al.*, 2001; Kerfant *et al.*, 2001). Additionally, microtubules appear to negatively regulate contractile activity and their levels are up-regulated in some models of cardiac hypertrophy (Klein, 1983; Tsutsui *et al.*, 1993). It is tempting to speculate that MURF-3 and MURF-2 work together in muscle differentiation, perhaps by microtubule stabilizing and titin-based signaling mechanisms (see below for discussion). Future investigations are required to address these important issues.

MURF-2: A link between multiple cytoskeletal arrays early in muscle development?

To date, only a few studies have explored the biochemical and functional properties of MURF-2, the only MURF family member that is developmentally down-regulated. However, these studies have begun to unravel a surprisingly complex story. The initial identifica-

tion of MURF-2 reported two MURF-2 isoforms generated by alternative splicing events in the 3' region with predicted sizes of 50 and 60 kDa (Centner *et al.*, 2001). Subsequent studies suggested the existence of two additional MURF-2 isoforms including: a small (~27 kDa), cardiac-specific variant missing the coiled-coil domains; and another 60 kDa isoform generated via alternative reading frame usage in the terminal exon (Pizon *et al.*, 2002; Lange *et al.*, 2005). Western blot analyses using two different polyclonal antibodies generated against the C-terminal region of MURF-2 (predicted to be present in all the isoforms) detected two abundant species that migrated at ~55 and ~60 kDa in embryonic heart and skeletal muscle lysates (McElhinny *et al.*, 2004). Other significant differences in the literature concerning MURF-2 include studies that did not detect a direct biochemical interaction between MURF-2 and titin (Centner *et al.*, 2001) and those that did (Pizon *et al.*, 2002). Further studies, with additional MURF-2-isoform specific reagents, are required to (1) clarify the isoform profiles in cardiac and skeletal muscles; (2) address whether the levels of its different isoforms change upon developmental down-regulation of MURF-2; and (3) establish whether MURF-2 interacts directly and/or indirectly with M-line titin.

Immunofluorescent microscopy studies in the C57 differentiating skeletal myocyte cell line revealed an intriguing, transient localization pattern of MURF-2 during myofibrillogenesis (Pizon *et al.*, 2002). First, like MURF-3, MURF-2 was found to colocalize with microtubules, with the exclusion of dynamic tyrosinated microtubules (Tyr-MTs); this staining pattern was observed early in myofibrillogenesis. Next, the MURF-2 labeled microtubules transiently associated with myosin and then A-band titin, and finally disappeared from the sarcomere upon the formation of mature myofibrils. Thus, it was proposed that MURF-2 functions as an “adaptor” to transiently link microtubules, titin and myosin during myofibril assembly, perhaps allowing the microtubules to act as a scaffold for immature myofibrils (Pizon *et al.*, 2002). In primary cultures of rat cardiac myocytes, MURF-2 was detected mainly diffuse in the cytoplasm, but also colocalized with a portion of microtubules and at the M-line region of the sarcomere (which contains the titin-binding site for MURF-1) (McElhinny *et al.*, 2004). Other studies also reported a change in MURF-2's localization from the M-line to nuclei in cultures of serum-starved cardiac myocytes and upon denervation of skeletal muscle in vivo (Pizon *et al.*, 2002; Lange *et al.*, 2005). These studies indicate that MURF-2 exhibits multiple localization patterns in muscle: future studies are needed to identify the specific MURF-2 isoforms that are targeted to these distinct locations in the cell.

Functional studies in cardiac myocytes revealed that a reduction of MURF-2 levels via antisense oligonucleotides dramatically perturbed the structure of Glu-MTs, and appeared to secondarily affect the integrity

of the distinct, but overlapping, population of stable acetylated microtubules (Ac-MTs). In contrast, dynamic Tyr-MTs were unaffected in their structure (McElhinny *et al.*, 2004). Thus, these data are consistent with the idea of MURF-2 functioning similarly to MURF-3 as a Glu-MT stabilizer in early developmental stages.

Interestingly, a known function of Glu-MTs is to regulate the localization and maintenance of intermediate filament proteins in many non-muscle cells (Gyoeva and Gelfand, 1991; Gurland and Gundersen, 1995). Therefore, it was next determined that a portion of both desmin and vimentin intermediate filaments colocalized with Glu-MTs in cardiac myocytes, and their integrity also appeared disrupted upon MURF-2 knock-down (McElhinny *et al.*, 2004). Whether this phenotype occurred secondarily to the Glu-MT perturbation remains an interesting question. Additionally, since MURF-2 assembles at the M-line, sarcomeric components were examined by immunofluorescence microscopy after its knock-down, and it was determined that the M-line region of titin and myomesin were specifically perturbed, but not other regions of the sarcomere including the Z lines, thin and thick filaments. Thus, MURF-2 also plays a role in the structure of the M-line, although the mechanism responsible for this is unknown. Not surprisingly, proper MURF-2 expression also was required for normal contractile activity of cardiac myocytes. Additionally, knock-down of MURF-2 in primary cultures of differentiating chick myotubes revealed that fusion events, as well as myofibrillogenesis, were delayed; this phenotype is similar to the one observed upon reduction of MURF-3 in C2C12 myotubes (Spencer *et al.*, 2000). Whether this is a result of the perturbed microtubule structure observed in the developing myotubes with reduced MURF-2 levels, or whether MURF-2 affects downstream events/pathways involved in myogenesis, remains to be determined.

Finally, a recent study suggests that MURF-2 is a key player in a signaling complex involving the titin kinase domain (Lange *et al.*, 2005). In this model, the zinc-finger protein, nbr1, interacts with the titin kinase domain in a conformation that appears to be mechanically induced. Nbr1 then binds to the ubiquitin-associated molecule, p62 (SQSTM1, ZIP or ORCA) that in turn interacts with MURF-2. The investigators also report that a mutation in the titin kinase domain disrupts this complex, resulting in muscle disease (Lange *et al.*, 2005). Further investigations into this pathway are required to assess whether MURF-2 indeed is involved in the regulation of sarcomeric protein turn-over and muscle gene expression via the titin kinase domain.

Conclusions

An increasing number of titin-associated proteins, including the MURFs, are being discovered that do

not fit the more traditional view of the sarcomere as a static scaffold (Sanger and Sanger, 2001). In the few years since their discovery, the MURF proteins have emerged as important players in many critical aspects of striated muscle differentiation, structure and function, and it is likely that additional roles for these proteins will be identified in the near future. These future endeavors, as well as further investigations into a few potentially conflicting observations already reported for some MURF localization patterns and associations (e.g., MURF-2 and -3), will undoubtedly depend on the generation of new powerful reagents including: isoform-specific antibodies; the characterization of the complete profile of MURF isoforms in different muscles; and thorough analyses of single, double and/or triple conditional MURF knock-out models. It will also be interesting to determine whether MURF or MURF-like proteins are present in invertebrates as well, and whether they share some of the same functions as their mammalian counterparts.

Importantly, although homologous in structure, the MURF family members differ in several properties including: distinct expression patterns during pre- and postnatal cardiac and skeletal muscle development, isoform profiles, binding partners, subcellular localizations and functions (for summary see Table 1). Many observations, especially the members' ability to hetero-oligomerize and their apparent (although perhaps transient) shared localization at the M-line region of titin, indicate the exciting possibility that all MURF family members may be regulated by titin-based mechanisms, either directly or indirectly. Specifically, the association of MURF-1 with its multiple binding partners and with M-line titin may link the giant protein, and perhaps its kinase domain, to ubiquitination pathways, nuclear transport, and/or gene expression. These processes appear to have critical implications in the physiology of skeletal muscle atrophy and in cardiomyocyte hypertrophy. For the other family members, differential expression and heterodimerization of MURF-2 and -3 may link together sarcomeres (via the titin kinase domain) with microtubule function and/or signaling pathways in striated muscles at different developmental stages. Furthermore, the identification of MURF-2 and -3 as mediators of the stable microtubule network, and as myogenic regulators, reveals another potential link between titin, microtubule organization and myogenesis.

Striated muscle structure has been studied for more than a century. Yet the identification and initial characterization of the MURFs (as well as other titin-binding proteins) underscore how much more investigation is required to decipher the functional properties of striated muscle. For example, advances in genetics, biochemistry, and molecular and cell biology, are now establishing entire new fields of study, including those seeking to identify the signaling mechanisms that myocytes require to differentiate, respond to biomechanical

and atrophy stimuli, and mechanisms that allow them to maintain their intricate structure. A plethora of immediate questions concerning the mechanisms of action of the MURFs need to be addressed including: Do MURF-2 and MURF-3 also have ubiquitin ligase activity, since they are highly homologous to MURF-1? If so, what are their targets and how are they regulated? In this regard, yeast 2-hybrid studies reveal MURF-1 and MURF-2 have some redundant binding partners (Witt *et al.*, 2005) (see Table 1). Do other MURFs also play a role in cardiac hypertrophy? Do MURF-2 and -3 interact directly with titin? Does MURF-1 also interact with microtubules? What proteins interact with MURF-1 and MURF-3 at the Z line? Do MURF-1, MURF-2 and/or MURF-3 have roles in transcriptional regulation? Investigations into the functional properties of the MURFs are exciting areas of research that will undoubtedly lead to fundamental findings about striated muscle development, structure and function.

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References

- Antin PB, Forry-Schaudies S, Friedman TM, Tapscott SJ and Holtzer H (1981) Taxol induces postmitotic myoblasts to assemble interdigitating microtubule-myosin arrays that exclude actin filaments. *J Cell Biol* **90**: 300–308.
- Arya R, Kedar V, Hwang JR, McDonough H, Li H-H and Taylor J (2004) Muscle ring finger protein-1 inhibits PKC ϵ activation and prevents cardiomyocyte hypertrophy. *J Cell Biol* **167**: 1147–1159.
- Bang ML, Centner T, Fornoff F, Geach A, Gotthardt M, McNabb M, Witt C, Labeit D, Gregorio CC, Granzier H and Labeit S (2001) The complete gene sequence of titin, expression of an unusual ~700 kDa titin isoform and its interaction with obscurin identify a novel Z-line to I-band linking system. *Circ Res* **89**: 1065–1072.
- Belmadani S, Pous C, Ventura-Clapier R, Fischmeister R and Mery PF (2002) Post-translational modifications of cardiac tubulin during chronic heart failure in the rat. *Mol Cell Biochem* **237**: 39–46.
- Benian GM, Kiff JE, Neckelmann N, Moerman DG and Waterston RH (1989) Sequence of an unusually large protein implicated in regulation of myosin activity in *C. elegans*. *Nature* **342**: 45–50.
- Bodine SC, Latres E, Baumhueter S, Lai VK, Nunez L, Clarke BA, Poueymirou WT, Panaro FJ, Na E, Dharmarajan K, Pan ZQ, Valenzuela DM, DeChiara TM, Stitt TN, Yancopoulos GD and Glass DJ (2001) Identification of ubiquitin ligases required for skeletal muscle atrophy. *Science* **294**: 1704–1708.
- Borden KL (1998) RING fingers and B-boxes: zinc-binding protein-protein interaction domains. *Biochem Cell Biol* **76**: 351–358.
- Borden KL (2000) RING domains: master builders of molecular scaffolds? *J Mol Biol* **295**: 1103–1112.
- Bullard B, Linke WA and Leonard K (2002) Varieties of elastic protein in invertebrate muscles. *J Muscle Res Cell Motil* **23**: 435–447.
- Cai D, Frantz JD, Tawa NE, Melendez PA, Oh BC, Lidov HG, hasselgren PO, Frontera WR, Lee J, Glass DJ and Shoelson SE (2004) IKK β /NF- κ B activation causes severe muscle wasting in mice. *Cell* **119**: 285–298.
- Calaghan SC, Le Guennec JY and White E (2001) Modulation of Ca²⁺ signaling by microtubule disruption in rat ventricular myocytes and its dependence on the ruptured patch-clamp configuration. *Circ Res* **88**: E32–E37.
- Centner T, Yano J, Kimura E, McElhinny AS, Pelin K, Witt CC, Bang ML, Trombitas K, Granzier H, Gregorio CC, Sorimachi H and Labeit S (2001) Identification of muscle specific ring finger proteins as potential regulators of the titin kinase domain. *J Mol Biol* **306**: 717–726.
- Chang W, Webster DR, Salam AA, Gruber D, Prasad A, Eiserich JP and Bulinski JC (2002) Alteration of the C-terminal amino acid of tubulin specifically inhibits myogenic differentiation. *J Biol Chem* **277**: 30690–30698.
- Dai KS and Liew CC (2001) A novel human striated muscle RING zinc finger protein, SMRZ, interacts with SMT3b via its RING domain. *J Biol Chem* **276**: 23992–23999.
- Dehoux MJ, van Beneden RP, Fernandez-Celemin L, Lausé PL and Thissen JP (2003) Induction of MafBx and Murf ubiquitin ligase mRNAs in rat skeletal muscle after LPS injection. *FEBS Lett* **544**: 214–217.
- Flaherty DB, Gernert KM, Shmeleva N, Tang X, Mercer KB, Borodovsky M and Benian GM (2002) Titins in *C. elegans* with unusual features: coiled-coil domains, novel regulation of kinase activity and two new possible elastic regions. *J Mol Biol* **323**: 533–549.
- Franch HA and Price SR (2005) Molecular signaling pathways regulating muscle proteolysis during atrophy. *Curr Opin Clin Nutr Metab Care* **8**: 271–275.
- Freemont PS (2000) RING for destruction? *Curr Biol* **10**: R84–R87.
- Gill G (2004) SUMO and ubiquitin in the nucleus: different functions, similar mechanisms? *Genes Dev* **18**: 2046–2059.
- Gomez AM, Kerfant BG and Vassort G (2000) Microtubule disruption modulates Ca(2+) signaling in rat cardiac myocytes. *Circ Res* **86**: 30–36.
- Gotthardt M, Hammer RE, Hubner N, Monti J, Witt CC, McNabb M, Richardson JA, Granzier H, Labeit S and Herz J (2003) Conditional expression of mutant M-line titins results in cardiomyopathy with altered sarcomere structure. *J Biol Chem* **278**: 6059–6065.
- Granzier H and Labeit S (2002) Cardiac titin: an adjustable multifunctional spring. *J Physiol* **541**: 335–342.
- Granzier HL and Labeit S (2004) The giant protein titin: a major player in myocardial mechanics, signaling, and disease. *Circ Res* **94**: 284–295.
- Gregorio CC and Fowler VM (1995) Mechanisms of thin filament assembly in embryonic chick cardiac myocytes: Tropomodulin requires tropomyosin for assembly. *J Cell Biol* **129**: 683–695.
- Gregorio CC, Trombitas K, Centner T, Komerer B, Stier G, Kunke K, Suzuki K, Obermayr F, Hermann B, Granzier H, Sorimachi H and Labeit S (1998) The NH2 terminus of titin spans the Z-disc: its interaction with a novel 19-kDa ligand (T-cap) is required for sarcomeric integrity. *J Cell Biol* **143**: 1013–1027.
- Gundersen GG, Khawaja S and Bulinski JC (1989) Generation of a stable, posttranslationally modified microtubule array is an early event in myogenic differentiation. *J Cell Biol* **109**: 2275–2288.
- Gurland G and Gundersen GG (1995) Stable, detyrosinated microtubules function to localize vimentin intermediate filaments in fibroblasts. *J Cell Biol* **131**: 1275–1290.

- Gustafson TA, Bahl JJ, Markham BE, Roeske WR and Morkin E (1987) Hormonal regulation of myosin heavy chain and alpha-actin gene expression in cultured fetal rat heart myocytes. *J Biol Chem* **262**: 13316–13322.
- Gyoeva FK and Gelfand VI (1991) Coalignment of vimentin intermediate filaments with microtubules depends on kinesin. *Nature* **353**: 445–448.
- Ikeda K, Emoto N, Matsuo M and Yokoyama M (2003) Molecular identification and characterization of a novel nuclear protein whose expression is up-regulated in insulin-resistant animals. *J Biol Chem* **278**: 3514–3520.
- Jeyaseelan R, Poizat C, Baker RK, Abdishoo S, Isterabadi LB, Lyons GE and Kedes L (1997) A novel cardiac-restricted target for doxorubicin. CARP, a nuclear modulator of gene expression in cardiac progenitor cells and cardiomyocytes. *J Biol Chem* **272**: 22800–22808.
- Kano Y, Fujimaki N and Ishikawa H (1991) The distribution and arrangement of microtubules in mammalian skeletal muscle fibers. *Cell Struct Funct* **16**: 251–261.
- Kedar V, McDonough H, Arya R, Li H-H, Rockman HA and Patterson C (2004) Muscle-specific RING finger 1 is a bona fide ubiquitin ligase that degrades cardiac troponin I. *Proc Natl Acad Sci, USA* **101**: 18135–18140.
- Kemp TJ, Sadusky TJ, Saltis F, Carey N, Moss J and Yang SY (2000) Identification of Ankrd2, a novel skeletal muscle gene coding for a stretch-responsive ankyrin-repeat protein. *Genomics* **66**: 229–241.
- Kerfant BG, Vassort G and Gomez AM (2001) Microtubule disruption by colchicine reversibly enhances calcium signaling in intact rat cardiac myocytes. *Circ Res* **88**: E59–E65.
- Klein I (1983) Colchicine stimulates the rate of contraction of heart cells in culture. *Cardiovasc Res* **17**: 459–465.
- Labeit S and Kolmerer B (1995) Titins, giant proteins in charge of muscle ultrastructure and elasticity. *Science* **270**: 293–296.
- Labeit S, Gautel M, Lakey A and Trinick J (1992) Towards a molecular understanding of titin. *EMBO J* **11**: 1711–1716.
- Lange S, Xiang F, et al. (2005) The kinase domain of titin controls muscle gene expression and protein turnover. Science: ScienceExpress.
- Latres E, Amini AR, Amini AA, Griffiths J, Martin FJ, Wei Y, Lin HC, Yancopoulos GD and Glass DJ (2005) IGF-1 inversely regulates atrophy-induced genes via the phosphatidylinositol 3-kinase/Akt/mammalian target of rapamycin (P13K/Akt/TOR) pathway. *J Biol Chem* **280**: 2737–2744.
- Lecker SH (2003) Ubiquitin-protein ligases in muscle wasting: multiple parallel pathways? *Curr Opin Clin Nutr Metab Care* **6**: 271–275.
- Lecker SH, Jagoe RT, Gilbert A, Gomes M, Baracos V, Bailey J, Price SR, Mitch WE and Goldberg AL (2004) Multiple types of skeletal muscle atrophy involve a common program of changes in gene expression. *FASEB J* **18**: 39–51.
- Linke WA and Granzier H (1998) A spring tale: new facts on titin elasticity. *Biophys J* **75**: 2613–2614.
- Maruyama K, Matsubara R, Natori Y, Nonomura S and Kimura S (1977) Connectin, an elastic protein of muscle. *J Biochem* **82**: 317–337.
- McElhinny AS, Kakinuma K, Sorimachi H, Labeit S and Gregorio CC (2002) Muscle-specific RING finger-1 interacts with titin to regulate sarcomeric M-line and thick filament structure and may have nuclear functions via its interaction with glucocorticoid modulatory element binding protein-1. *J Cell Biol* **157**: 125–136.
- McElhinny AS, Perry CN, Witt C, Labeit S and Gregorio CC (2004) Muscle specific RING finger-2 (MURF-2) has multiple subcellular localizations and is required for microtubule, intermediate filament, and sarcomeric M-line maintenance in striated muscle. *J Cell Sci* **117**: 3175–3188.
- Melchior F (2000) SUMO–nonclassical ubiquitin. *Annu Rev Cell Dev Biol* **16**: 591–626.
- Melchior F, Schergaut M and Pichler A (2003) SUMO: ligases, isopeptidases and nuclear pores. *Trends Biochem Sci* **28**: 612–618.
- Miller M, Bang ML, Witt CC, Labeit D, Trombitas C, Watanabe K, Granzier H, McElhinny AS, Gregorio CC and Labeit S (2003) The muscle ankyrin repeat proteins: CARP/ankrd2/DARP as a family of titin filament based stress response molecules. *J Mol Biol* **333**: 951–964.
- Miller MK, Granzier H, Ehler E and Gregorio CC (2004) The sensitive giant: the role of titin-based stretch sensing complexes in the heart. *Trends Cell Biol* **14**: 119–126.
- Mues A, van der Ven PF, Young P, Furst DO and Gautel M (1998) Two immunoglobulin-like domains of the Z-disc portion of titin interact in a conformation-dependent way with telethonin. *FEBS Lett* **428**: 111–114.
- Nikawa T, Ishidoh K, Hirasaka K, Ishihara I, Ikemoto M, Kano M, Kominami E, Nonaka I, Ogawa T, Adams GR, Baldwin KM, Yasui N, Kishi K and Takeda S (2004) Skeletal muscle gene expression in space-flown rats. *FASEB J* **18**: 522–524.
- Obermann WM, Gautel M, Weber K and Fürst DO (1997) Molecular structure of the sarcomeric M band: mapping of titin and myosin binding domains in myomesin and the identification of a potential regulatory phosphorylation site in myomesin. *EMBO J* **16**: 211–220.
- Olson NJ, Pearson RB, Needleman DS, Hurwitz MY, Kemp BE and Means AR (1990) Regulatory and structural motifs of chicken gizzard myosin light chain kinase. *Proc Natl Acad Sci USA* **87**: 2284–2288.
- Perhonen M, Sharp WW and Russell B (1998) Microtubules are needed for dispersal of alpha-myosin heavy chain mRNA in rat neonatal cardiac myocytes. *J Mol Cell Cardiol* **30**: 1713–1722.
- Pizon V, Iakovenko A, van Der Ven PF, Kelly R, Fatu C, Furst DO, Karsenti E and Gautel M (2002) Transient association of titin and myosin with microtubules in nascent myofibrils directed by the MURF2 RING-finger protein. *J Cell Sci* **115**: 4469–4482.
- Sacheck JM, Ohtsuka A, McLary SC and Goldberg AL (2004) IGF-1 stimulates muscle growth by suppressing protein breakdown and expression of atrophy-related ubiquitin ligases, atrogin-1 and MURF1. *Am J Physiol Endocrinol Metab* **287**: E591–E601.
- Saitoh O, Arai T and Obinata T (1988) Distribution of microtubules and other cytoskeletal filaments during myotube elongation as revealed by fluorescence microscopy. *Cell Tissue Res* **252**: 263–273.
- Saitoh H and Hinchey J (2000) Functional heterogeneity of small ubiquitin-related protein modifiers SUMO-1 versus SUMO-2/3. *J Biol Chem* **275**: 6252–6258.
- Sanger JW and Sanger JM (2001) Fishing out proteins that bind to titin. *J Cell Biol* **154**: 21–24.
- Sorimachi H, Kinbara K, Kimura S, Takahashi M, Ishiura S, Sasagawa N, Sorimachi N, Shimada H, Tagawa K and Maruyama K (1995) Muscle-specific calpain responsible for limb girdle muscular dystrophy type 2A, associates with connectin through IS2, a p94-specific sequence. *J Biol Chem* **270**: 31158–31162.
- Southgate A, Vigoreaux J, Benian G and Pardue ML (1991) Drosophila has a twitchin/titin-related gene that appears to encode projectin. *Proc Natl Acad Sci USA* **88**: 7973–7977.
- Spencer JA, Eliazar S, Ilaria RL, Richardson JA Jr and Olson EN (2000) Regulation of microtubule dynamics and myogenic differentiation by MURF, a striated muscle RING-finger protein. *J Cell Biol* **150**: 771–784.
- Stitt TN, Drujan D, Clarke BA, Panaro F, Timofeyeva Y, Kline WO, Gonzalez M, Yancopoulos GD and Glass DJ (2004) The IGF-1/P13K/Akt pathway prevents expression of muscle atrophy-induced ubiquitin ligases by inhibiting FOXO transcription factors. *Mol Cell* **14**: 395–403.
- Takahashi M, Tsutsui H, Tagawa H, Igarashi-Saito K, Imanaka-Yoshida K and Takeshita A (1998) Microtubules are involved in early hypertrophic responses of myocardium during pressure overload. *Am J Physiol* **275**: H341–H348.
- Toyama Y, Forry-Schaudies S, Hoffman B and Holtzer H (1982) Effects of taxol and Colcemid on myofibrillogenesis. *Proc Natl Acad Sci USA* **79**: 6556–6560.
- Tskhovrebova L and Trinick J (2003) Titin: properties and family relationships. *Nat Rev Mol Cell Biol* **4**: 679–689.
- Tskhovrebova L and Trinick J (2004) Properties of titin immunoglobulin and fibronectin-3 domains. *J Biol Chem* **279**: 46351–46354.

- Tsutsui H, Ishihara K and Cooper GT (1993) Cytoskeletal role in the contractile dysfunction of hypertrophied myocardium. *Science* **260**: 682–687.
- Wang K., McClure J and Tu A (1979) Titin: major myofibrillar components of striated muscle. *Proc Natl Acad Sci USA*. **76**: 3698–3702.
- Webster DR (1997) Regulation of post-translationally modified microtubule populations during neonatal cardiac development. *J Mol Cell Cardiol* **29**: 1747–1761.
- Witt SH, Granzier H, Witt CC and Labeit S (2005) MURF-1 and MURF-2 target a specific subset of myofibrillar proteins redundantly: towards understanding MURF-dependent muscle ubiquitination. *J Mol Biol* **350**: 713–722.
- Young P, Ehler E and Gautel M (2001) Obscurin, a giant sarcomeric Rho guanine nucleotide exchange factor protein involved in sarcomere assembly. *J Cell Biol* **154**: 123–136.
- Zou Y, Evans S, Chen J, Kuo HC, Harvey RP and Chien KR (1997) CARP, a cardiac ankyrin repeat protein, is downstream in the Nkx2–5 homeobox gene pathway. *Development* **124**: 793–804.